Chapter 8

Linear Transformations

Homework list

Strang

Sec 8.1:3, 10, 12, 15

Sec 8.2: 1, 2, 3, 10, 14, 15, 24, 26

Sec 8.3: 2

Homework

- 1. The Fourier Basis is a commonly used orthogonal basis for function space, given by:

$$\{1, \sin x, \cos x, \sin 2x, \cos 2x, \cdots\}$$

(a) Find the matrix representation of the derivative operator $D = \left[\frac{d}{dx}\right]$ relative to this basis.

(b) Find the matrix representation of the integral operator $S = \left[\int dx\right]$ relative to this basis.

(c) Evaluate DS and SD, and discuss your observations.

- 2. The Legendre Polynomials are an orthogonal basis for polynomials over the interval $x \in [-1, 1]$, For P^3, they are:

$$\left\{1, x, x^2 - \frac{1}{3}, x^3 - \frac{3}{5}x\right\}$$

Find the coordinate vector for $p(x) = 5x^3 - 3x^2 - x$ relative to this basis.

Section 8.1 & 8.2

The idea and matrix of a linear transformation

Review: Basis and dimension of $\mathbb{R}^{m\times n}$

- **Each basis "vector" must be an m-by-n matrix; one basis is the set of matrices E_{ij} with a single "1" at position ij and zeros everywhere else.**
- **There are $m \times n$ matrices of this type, so $\dim(\mathbb{R}^{m\times n}) = mn$.**

- **Using $\mathbb{R}^{2\times 3}$ as an example:**

$$\left\{\begin{bmatrix}1&0&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&1&0\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&1\\0&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\1&0&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&1&0\end{bmatrix}, \begin{bmatrix}0&0&0\\0&0&1\end{bmatrix}\right\}$$

  - **The set is obviously linear independent.**
  - **The set is a spanning set:**

$$\begin{bmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\end{bmatrix} = a_{11}\begin{bmatrix}1&0&0\\0&0&0\end{bmatrix} + a_{12}\begin{bmatrix}0&1&0\\0&0&0\end{bmatrix} + a_{13}\begin{bmatrix}0&0&1\\0&0&0\end{bmatrix} + a_{21}\begin{bmatrix}0&0&0\\1&0&0\end{bmatrix} + a_{22}\begin{bmatrix}0&0&0\\0&1&0\end{bmatrix} + a_{23}\begin{bmatrix}0&0&0\\0&0&1\end{bmatrix}$$

  - **The dimension is 6.**

Polynomial space.

- **Consider the set of all functions of a single variable that can be written as a polynomial of degree n or less.:**

$$P^n = \{ f(x) \mid f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n \}$$

- **When we add polynomials, we get another polynomial of no greater degree:**

$$f(x) + g(x) \in P^n, \forall f(x), g(x) \in P^n$$

- **When we scale polynomials, we get another polynomial of the same degree:**

$$kf(x) \in P^n, \forall f(x) \in P^n, \forall k \in \mathbb{R}$$

- **The set of polynomials is closed with respect to addition and scalar multiplication.**
- **The set of all polynomial functions of a single variable is a vector space; it is a subspace of function space!**

Basis and dimension of P^n

- **Each basis "vector" must be a polynomial of degree n or less; one basis is the set of functions $\{x^i, \quad i = [0, n]\}$.**
- **There are $n + 1$ functions of this type, so $\dim(P^n) = n + 1$.**

- **Using P^3 as an example:**

$$\{1, \quad x, \quad x^2, \quad x^3\}$$

  - **The set is obviously linear independent.**
  - **The set is a spanning set:**

$$p(x) = a_0(1) + a_1 x + a_2 x^2 + a_3 x^3$$

  - **The dimension is 4.**

Basis and dimension of function space?

- Any continuous function on a closed domain can be uniformly approximated as closely as desired by a polynomial (Stone-Weierstrass Theorem).
- **But there is no bound on the degree!**
- What is the dimension of the vector space of polynomials of ***any*** degree?
- Basis:

$$\{1, \quad x, \quad x^2, \quad x^3, \quad \cdots\}$$

- This set is countable but infinite (just like the natural numbers).

$$\dim(P) = \infty$$

  - Formally, $\dim(P) = \aleph_0$, the cardinality of the natural numbers
- In many applications, we approximate function space by using a finite basis (for example, polynomials up to degree n).

What is a linear map or linear transformation?

- **A linear map or linear transformation is a rule that uniquely converts one vector (of any type) into another, and which preserves linear combinations.**
- **Consider a rule to convert a vector from vector space U to one from vector space V, $F: V \to U$.**
- **To be a linear map, the rule must satisfy:**

$$\forall \vec{v} \in V, \exists! \, \vec{u} \in U: \vec{u} = F(\vec{v})$$

$$F(k\vec{v}_1 + l\vec{v}_2) = kF(\vec{v}_1) + lF(\vec{v}_2), \qquad \forall \vec{v}_1, \vec{v}_2 \in V$$

Special Facts:

- Identity transformation: F(0) = 0 for all linear transformations.
- Shift is not a linear transformation.

$$T(\boldsymbol{v}) = \boldsymbol{v} + \boldsymbol{u}_0 \text{ and } T(\boldsymbol{w}) = \boldsymbol{w} + \boldsymbol{u}_0.$$

Shift is not linear $\quad \boldsymbol{v} + \boldsymbol{w} + \boldsymbol{u}_0 \quad$ is not $\quad T(\boldsymbol{v}) + T(\boldsymbol{w}) = (\boldsymbol{v} + \boldsymbol{u}_0) + (\boldsymbol{w} + \boldsymbol{u}_0).$

- Dot products are linear.

Example 1 Choose a fixed vector $\boldsymbol{a} = (1, 3, 4)$, and let $T(\boldsymbol{v})$ be the dot product $\boldsymbol{a} \cdot \boldsymbol{v}$:

The input is $\boldsymbol{v} = (v_1, v_2, v_3)$. The output is $T(\boldsymbol{v}) = \boldsymbol{a} \cdot \boldsymbol{v} = v_1 + 3v_2 + 4v_3$.

Ex 2:

Question: Is the length T(v) = ||v|| a linear mapping?

Answer: No!!! $T(cv) \neq cT(v)(c = -1)\ and\ T(v + w) \neq T(v) + T(w)(Triagnular\ inequality)$.

Question 3: Is a rotation that rotates every vector by 30 degrees a linear map?

Answer: Yes. The sum of rotations T(v) + T(w) is the same as the rotation T(v + w) of the sum. The whole plane is turning together, in this linear transformation.

Generalization of linearity into more vectors

The rule of linearity extends to combinations of three vectors or n vectors:

Linearity $\boldsymbol{u} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_n\boldsymbol{v}_n$ **must transform to**

$$T(\boldsymbol{u}) = c_1T(\boldsymbol{v}_1) + c_2T(\boldsymbol{v}_2) + \cdots + c_nT(\boldsymbol{v}_n) \tag{1}$$

- Fact: You know T(u) for every vector u in the space if you know $T(v)\ for$ every $basis\ vector\ v_i\ in\ the\ space$.

Kernel and Range

- **The kernel of a linear map is the set of input values that map to zero:**

$$kernel(F: V \to U) = \{\vec{v} \in V: F(\vec{v}) = \vec{0}\}$$

  - **Comparable to the null space of a matrix.**

- **The range of a linear map is the set of possible output values:**

$$range(F: V \to U) = \{\vec{u} \in U: \exists \vec{v} \in V, \vec{u} = F(\vec{v})\}$$

  - **Comparable to the column space of a matrix.**

Matrix multiplication is a linear map!

- **Consider an m-by-n matrix A:**

$$\forall \vec{v} \in \mathbb{R}^n, \exists! \vec{u} \in \mathbb{R}^m : \vec{u} = A\vec{v}$$

$$A(k\vec{v}_1 + l\vec{v}_2) = kA\vec{v}_1 + lA\vec{v}_2, \qquad \forall \vec{v}_1, \vec{v}_2 \in \mathbb{R}^n$$

- $F(\vec{v}) = A\vec{v}$ **is linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$.**
- **In this case, the kernel of F is identical to the null space of A, and the range of F is identical to the column space of A.**

Derivatives are linear maps!

- **Consider the set of all smooth (infinitely differentiable) functions of a single variable, C^∞:**

$$\forall f(x) \in C^\infty, \exists! f'(x) \in C^\infty: f'(x) = \frac{d}{dx} f(x)$$

$$\frac{d}{dx}\big(kf(x) + lg(x)\big) = k\frac{d}{dx}f(x) + l\frac{d}{dx}g(x) = kf'(x) + lg'(x), \qquad \forall f(x), g(x) \in C^\infty$$

Example: Transformation F= du/dx

Find the null & column space of F(u) = du/dx for u= a + $bx + cx^2$.

Nullspace of $T(u) = du/dx$. For the nullspace we solve $T(u) = 0$. The derivative is zero when u *is a constant function*. So the one-dimensional nullspace is a line in function space—all multiples of the special solution $u = 1$.

Column space of $T(u) = du/dx$. In our example the input space contains all quadratics $a + bx + cx^2$. The outputs (the column space) are all linear functions $b + 2cx$. Notice that the **Counting Theorem** is still true: $\boldsymbol{r} + (\boldsymbol{n} - \boldsymbol{r}) = \boldsymbol{n}$.

dimension (**column space**) + dimension (**nullspace**) = **2** + **1** = **3** = dimension (**input space**)

Linear maps can be represented by matrices!

coefficients

- Coordinate vectors link generalized vector spaces with ordered lists:

$$V = span(\{\vec{v}_1, \quad \vec{v}_2, \quad \cdots \quad \vec{v}_n\}) \rightarrow \vec{x} = \sum_{i=1}^{n} a_i \vec{v}_i \,, \forall \vec{x} \in V \rightarrow [\vec{x}] = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

- Consider a linear map, $F: V \rightarrow U$:

$$F(\vec{x}) = F\left(\sum_{i=1}^{n} a_i \vec{v}_i\right) = \sum_{i=1}^{n} a_i F(\vec{v}_i) \,, \forall \vec{x} \in V$$

- The column view of a linear system gives:

$$\sum_{i=1}^{n} a_i F(\vec{v}_i) = [F(\vec{v}_1) \quad F(\vec{v}_2) \quad \cdots \quad F(\vec{v}_n)] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = [F][\vec{x}]$$

- The matrix representation of a linear map is a matrix whose columns are the mapped basis!

Linear maps can be represented by matrices!

- **Coordinate vectors link generalized vector spaces with ordered lists:**

$$V = span(\{\vec{v}_1, \quad \vec{v}_2, \quad \cdots \quad \vec{v}_n\}) \rightarrow \vec{x} = \sum_{i=1}^{n} a_i \vec{v}_i \,, \forall \vec{x} \in V \rightarrow [\vec{x}] = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

- **Consider a linear map,** $F: V \rightarrow U$:

$$F(\vec{x}) = F\left(\sum_{i=1}^{n} a_i \vec{v}_i\right) = \sum_{i=1}^{n} a_i F(\vec{v}_i) \,, \forall \vec{x} \in V$$

- $F(\vec{x})$ **can also be represented as a coordinate vector:**

$$U = span(\{\vec{u}_1, \quad \vec{u}_2, \quad \cdots \quad \vec{u}_n\}) \rightarrow F(\vec{x}) = \sum_{i=1}^{m} b_i \vec{u}_i \,, \forall \vec{x} \in V \rightarrow [F(\vec{x})] = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$F(\vec{x}) = \sum_{i=1}^{n} a_i F(\vec{v}_i) \quad \rightarrow \quad [F(\vec{x})] = \sum_{i=1}^{n} a_i [F(\vec{v}_i)]$$

Linear maps can be represented by matrices!

- **Coordinate vectors link generalized vector spaces with ordered lists:**

$$F(\vec{x}) = \sum_{i=1}^{n} a_i F(\vec{v}_i) \quad \rightarrow \quad [F(\vec{x})] = \sum_{i=1}^{n} a_i [F(\vec{v}_i)]$$

- **The column view of a linear system gives:**

$$\sum_{i=1}^{n} a_i [F(\vec{v}_i)] = [[F(\vec{v}_1)] \quad [F(\vec{v}_2)] \quad \cdots \quad [F(\vec{v}_n)]] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = [F][\vec{x}]$$

- **The matrix representation of a linear map is a matrix whose columns are the coordinate vectors of the mapped basis!**
- **When generalized vector spaces are written in terms of coordinate vectors, linear mappings are just matrix multiplication; another view of $A\vec{x} = \vec{b}$!**

$$[F(\vec{x})] = [F][\vec{x}] \leftrightarrow A\vec{x} = \vec{b}$$

How could find the matrix for this linear map?

Input u
$a + bx + cx^2$

Output $\dfrac{du}{dx} = b + 2cx.$

$$\boxed{\boldsymbol{A} = \begin{bmatrix} 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{2} \end{bmatrix} = \text{matrix form of the derivative } T = \frac{d}{dx}.}$$

Input u
$a + bx + cx^2$

Multiplication $Au = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} b \\ 2c \end{bmatrix}$

Output $\dfrac{du}{dx} = b + 2cx.$

The connection from T to A (we will connect every transformation to a matrix) depended on choosing an input basis $1, x, x^2$ and an output basis $1, x$.

Ex: The differential matrix for P^3

- **A simple basis of P^3:**

$$E = \{1, \quad x, \quad x^2, \quad x^3\}, f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \rightarrow \quad [f(x)]_E = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

- **The derivatives of the basis functions:**

$$e_1(x) = 1, e'_1(x) = 0 \rightarrow [e'_1(x)]_E = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, e_2(x) = x, e'_2(x) = 1 \rightarrow [e'_2(x)]_E = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$e_3(x) = x^2, e'_3(x) = 2x \rightarrow [e'_3(x)]_E = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}, e_4(x) = x^3, e'_4(x) = 3x^2 \rightarrow [e'_4(x)]_E = \begin{bmatrix} 0 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

Ex: The differential matrix for P^3

- **The differential matrix:**

$$D = \left[\frac{d}{dx}\right] = [[e'_1(x)]_E \quad [e'_2(x)]_E \quad [e'_3(x)]_E \quad [e'_4(x)]_E] = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{vmatrix}$$

- **We can find the derivative simply by performing matrix multiplication:**

$$D[f(x)]_E = \begin{vmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{vmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ 2a_2 \\ 3a_3 \\ 0 \end{bmatrix} = [f'(x)]_E \rightarrow f'(x) = a_1 + 2a_2x + 3a_3x^2$$

$$\frac{d}{dx}(a_0 + a_1x + a_2x^2 + a_3x^3) = a_1 + 2a_2x + 3a_3x^2$$

Integration T^+ is also linear!

Example 5 **Integration T^+ is also linear**: $\int_0^x (D + Ex)\,dx = Dx + \frac{1}{2}Ex^2$.

The input basis is now $1, x$. The output basis is $1, x, x^2$. The matrix A^+ for T^+ is 3 by 2:

Input v $D + Ex$ **Multiplication** $A^+v = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} D \\ E \end{bmatrix} = \begin{bmatrix} 0 \\ D \\ \frac{1}{2}E \end{bmatrix}$ **Output = Integral of** v $T^+(v) = Dx + \frac{1}{2}Ex^2$

Inverse of Linear Transformation

Example 7 Suppose A is an *invertible matrix*. Certainly $T(\boldsymbol{v}+\boldsymbol{w}) = A\boldsymbol{v} + A\boldsymbol{w} = T(\boldsymbol{v}) + T(\boldsymbol{w})$. Another linear transformation is multiplication by A^{-1}. This produces the ***inverse transformation*** T^{-1}, which brings every vector $T(\boldsymbol{v})$ back to $\boldsymbol{v}$:

$$T^{-1}(T(\boldsymbol{v})) = \boldsymbol{v} \quad \text{matches the matrix multiplication} \quad A^{-1}(A\boldsymbol{v}) = \boldsymbol{v}.$$

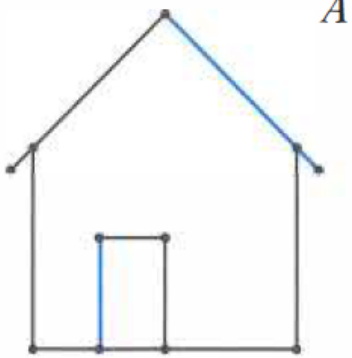
If $T(\boldsymbol{v}) = A\boldsymbol{v}$ and $S(\boldsymbol{u}) = B\boldsymbol{u}$, then the product $T(S(u))$ matches the product $AB\boldsymbol{u}$.

Note: A nonlinear transformation T is invertible if every b in the output space comes from exactly one x in the input space: T(x) = b always has exactly one solution.

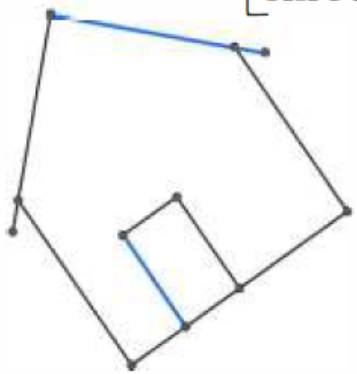
Ex:

House matrix $H = \begin{bmatrix} -6 & -6 & -7 & 0 & 7 & 6 & 6 & -3 & -3 & 0 & 0 & -6 \\ -7 & 2 & 1 & 8 & 1 & 2 & -7 & -7 & -2 & -2 & -7 & -7 \end{bmatrix}.$

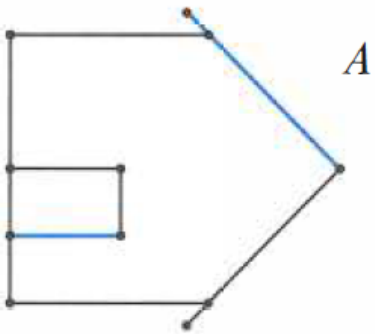
$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$



$A = \begin{bmatrix} \cos 35^\circ & -\sin 35^\circ \\ \sin 35^\circ & \cos 35^\circ \end{bmatrix}$



$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$



$A = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix}$

Figure 8.2: Linear transformations of a house drawn by **plot2d**$(A * H)$.

Sec 8.2

The matrix of a linear transformation

The matrix of a linear transformation

Assign a matrix A to every linear transformation T from $V = R^n$ to $W = R^m$. All vector spaces V and W have bases and each choice of those bases leads to matrix for T (the easiest base is the standard basis of unit vectors).

Suppose we know $T(v)$ for the input basis vectors v_1 to v_n.
Columns 1 to n of the matrix will contain those outputs $T(v_1)$ to $T(v_n)$.
A times c = matrix times vector = combination of those n columns.
Ac is the correct combination $c_1T(v_1) + \cdots + c_nT(v_n) = T(v)$.

Reason Every $\boldsymbol{v}$ is a unique combination $c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n$ of the basis vectors $\boldsymbol{v}_j$. Since T is a linear transformation (here is the moment for linearity), $T(\boldsymbol{v})$ must be **the same combination $c_1T(\boldsymbol{v}_1) + \cdots + c_nT(\boldsymbol{v}_n)$ of the outputs $T(\boldsymbol{v}_j)$ in the columns.**

Example

Question: Suppose a linear transformation T from R^2 to R^3 transforms $v_1 = (1,0)$ to $T(v_1) = (2,3,4)$;

$$v_2 = (0,1) \; to \; T(v_2) = (5,5,5).$$

a) What is the matrix for T?

b) What is Tc where c= (1, 1) .

$$A = \begin{bmatrix} 2 & 5 \\ 3 & 5 \\ 4 & 5 \end{bmatrix} \qquad c_1 = 1 \text{ and } c_2 = 1 \text{ give } T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = \begin{bmatrix} 2 & 5 \\ 3 & 5 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

Change of Basis:

Suppose the input space V and output space W are both R^2 . Suppose that T(v) = v is the identity transformation. What is the matrix for T?

The matrix is I only happens when the input basis is the same as the output basis.

When the input basis is in the columns of a matrix $\boldsymbol{V}$, and the output basis is in the columns of $\boldsymbol{W}$, the change of basis matrix for $T = I$ is $\boldsymbol{B} = \boldsymbol{W}^{-1}\boldsymbol{V}$.

This formula $\boldsymbol{B} = \boldsymbol{W}^{-1}\boldsymbol{V}$ produces one of the world's greatest mysteries: When the *standard basis* $\boldsymbol{V} = \boldsymbol{I}$ is changed to a different basis $\boldsymbol{W}$, **the change of basis matrix is not $\boldsymbol{W}$ but $\boldsymbol{B} = \boldsymbol{W}^{-1}$**. Larger basis vectors have smaller coefficients!

Key in change of basis

$$\begin{aligned} \boldsymbol{u} &= c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n \\ \boldsymbol{u} &= d_1\boldsymbol{w}_1 + \cdots + d_n\boldsymbol{w}_n \end{aligned} \quad \text{is} \quad \begin{bmatrix} \boldsymbol{v}_1 & \cdots & \boldsymbol{v}_n \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \boldsymbol{w}_1 & \cdots & \boldsymbol{w}_n \end{bmatrix} \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} \quad \text{and } \boldsymbol{V}\boldsymbol{c} = \boldsymbol{W}\boldsymbol{d}.$$

The coefficients $\boldsymbol{d}$ in the new basis of $\boldsymbol{w}$'s are $\boldsymbol{d} = \boldsymbol{W}^{-1}\boldsymbol{V}\boldsymbol{c}$. Then $\boldsymbol{B}$ is $\boldsymbol{W}^{-1}\boldsymbol{V}$. (2)

This formula $\boldsymbol{B} = \boldsymbol{W}^{-1}\boldsymbol{V}$ produces one of the world's greatest mysteries: When the *standard basis* $\boldsymbol{V} = \boldsymbol{I}$ is changed to a different basis $\boldsymbol{W}$, **the change of basis matrix is not $\boldsymbol{W}$ but $\boldsymbol{B} = \boldsymbol{W}^{-1}$**. Larger basis vectors have smaller coefficients!

Example

Question 1: Suppose u = $\begin{bmatrix} x \\ y \end{bmatrix}$ under the standard basis of R^2. What is the coefficient for u under $\{w_1, w_2\}$ basis?

$\begin{bmatrix} x \\ y \end{bmatrix}$ in the standard basis has coefficients $\begin{bmatrix} \boldsymbol{w}_1 & \boldsymbol{w}_2 \end{bmatrix}^{-1} \begin{bmatrix} x \\ y \end{bmatrix}$ in the $\boldsymbol{w}_1, \boldsymbol{w}_2$ basis.

Question 2: Suppose u = $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ under the standard basis of R^2. What is the coefficient for u under $\{\begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}\}$ basis?

Example:

Suppose the input space V and output space W are both R^2. Assume that T(v) = v is the identity transformation. What is the matrix for T if the input basis is $\{\begin{bmatrix}3\\3\end{bmatrix}, \begin{bmatrix}6\\8\end{bmatrix}\}$ and the output basis is $\{\begin{bmatrix}3\\1\end{bmatrix}, \begin{bmatrix}0\\2\end{bmatrix}\}$?

Input basis $\begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 3 & 8 \end{bmatrix}$ **Output basis** $\begin{bmatrix} w_1 & w_2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$ **Change of basis** $v_1 = 1w_1 + 1w_2$, $v_2 = 2w_1 + 3w_2$

Matrix B for change of basis $\begin{bmatrix} w_1 & w_2 \end{bmatrix} \begin{bmatrix} B \end{bmatrix} = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$ is $\begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 3 & 8 \end{bmatrix}$.

How could we construct a matrix for any linear transformation?

Step1 : Suppose T transforms the space V (n-dim) to the space W (m-dim). We choose a basis $v_1, v_2, \cdots v_n$ for V and we choose a basis $w_1, w_2, \cdots w_m$ for W. The matrix A will be m by n.

Step 2: To find the first column of A, apply T to the first basis vector v_1. The output T (v_1) is in W.

$T(\boldsymbol{v}_1)$ ***is a combination*** $a_{11}\boldsymbol{w}_1 + \cdots + a_{m1}\boldsymbol{w}_m$ ***of the output basis for*** $\mathbf{W}$.

These numbers $a_{11}, \ldots, a_{m1}$ *go into the first column of* A. Transforming $\boldsymbol{v}_1$ to $T(\boldsymbol{v}_1)$ matches multiplying $(1, 0, \ldots, 0)$ by A. It yields that first column of the matrix. When T is the derivative and the first basis vector is 1, its derivative is $T(\boldsymbol{v}_1) = \mathbf{0}$. So for the derivative matrix below, the first column of A is all zero.

Example

Example 3 The input basis of $\boldsymbol{v}$'s is $1, x, x^2, x^3$. The output basis of $\boldsymbol{w}$'s is $1, x, x^2$. Then $\boldsymbol{T}$ **takes the derivative**: $\boldsymbol{T}(\boldsymbol{v}) = \dfrac{d\boldsymbol{v}}{dx}$ and A = "derivative matrix".

If $\boldsymbol{v} = c_1 + c_2 x + c_3 x^2 + c_4 x^3$
then $\dfrac{d\boldsymbol{v}}{dx} = \mathbf{1}c_2 + \mathbf{2}c_3 x + \mathbf{3}c_4 x^2$

$$A\boldsymbol{c} = \begin{bmatrix} 0 & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{2} & 0 \\ 0 & 0 & 0 & \mathbf{3} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} c_2 \\ 2c_3 \\ 3c_4 \end{bmatrix}$$

Key rule: The jth column of A is found by applying T to the jth basis vector $\boldsymbol{v}_j$

$$T(\boldsymbol{v}_j) = \text{combination of output basis vectors} = a_{1j}\boldsymbol{w}_1 + \cdots + a_{mj}\boldsymbol{w}_m. \tag{3}$$

Example:

Question: For the integral $T^+(v): V(3-dim) \to W(4-dim)$, we choose $1, x, x^2$ as a basis for V and $1, x, x^2, x^3$ as a basis for W. What is the integral matrix A^+ for the linear transformation T^+?

The integral of $d_1 + d_2x + d_3x^2$

is $\quad d_1x + \frac{1}{2}d_2x^2 + \frac{1}{3}d_3x^3$

$$A^+d = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} 0 \\ d_1 \\ \frac{1}{2}d_2 \\ \frac{1}{3}d_3 \end{bmatrix}$$

Matrix products AB match transformations TS

When we apply the transformation T to the output from S, we get TS by this rule: $(TS)(\boldsymbol{u})$ *is defined to be* $\boldsymbol{T}(\boldsymbol{S}(\boldsymbol{u}))$. The output $S(\boldsymbol{u})$ becomes the input to T.

When we apply the matrix A to the output from B, we multiply AB by this rule: $(AB)(\boldsymbol{x})$ is defined to be $\boldsymbol{A}(\boldsymbol{B}\boldsymbol{x})$. The output $B\boldsymbol{x}$ becomes the input to A.
Matrix multiplication gives the correct matrix AB to represent TS.

Multiplication The linear transformation TS starts with any vector $\boldsymbol{u}$ in $\mathbf{U}$, goes to $S(\boldsymbol{u})$ in $\mathbf{V}$ and then to $T(S(\boldsymbol{u}))$ in $\mathbf{W}$. The matrix AB starts with any $\boldsymbol{x}$ in $\mathbf{R}^p$, goes to $B\boldsymbol{x}$ in $\mathbf{R}^n$ and then to $AB\boldsymbol{x}$ in $\mathbf{R}^m$. **The matrix AB correctly represents TS:**

$$TS: \quad \mathbf{U} \to \mathbf{V} \to \mathbf{W} \qquad\qquad AB: \quad (m \text{ by } n)(n \text{ by } p) = (m \text{ by } p).$$

Example:

Example 6 S rotates by the angle θ and T rotates by $-\theta$. Then $TS = I$ leads to $AB = I$.

In this case $T(S(\boldsymbol{u}))$ is $\boldsymbol{u}$. We rotate forward and back. For the matrices to match, $AB\boldsymbol{x}$ must be $\boldsymbol{x}$. *The two matrices are inverses*. Check this by putting $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$ into the backward rotation matrix A:

$$AB = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \cos^2\theta + \sin^2\theta \end{bmatrix} = I.$$

What is the best bases?

Best base: Choose transformation that diagonalize the matrix A.

- With the standard basis, the transformation produces a matrix that probably not diagonal.
- The two great choices are eigenvectors and singular vectors: Choose the bases that diagonalize the matrix:

Eigenvectors If T transforms $\mathbf{R}^n$ to $\mathbf{R}^n$, its matrix A is square. But using the standard basis, that matrix A is probably not diagonal. If there are n independent eigenvectors, *choose those as the input and output basis*. In this good basis, **the matrix for T is the diagonal eigenvalue matrix Λ.**

Example: Not diagonalizable

Example 7 **The projection matrix** T projects every $\boldsymbol{v} = (x, y)$ in $\mathbf{R}^2$ onto the line $y = -x$. Using the standard basis, $\boldsymbol{v}_1 = (1, 0)$ projects to $T(\boldsymbol{v}_1) = \left(\frac{1}{2}, -\frac{1}{2}\right)$. For $\boldsymbol{v}_2 = (0, 1)$ the projection is $T(\boldsymbol{v}_2) = \left(-\frac{1}{2}, \frac{1}{2}\right)$. Those are the columns of A:

Projection matrix
Standard bases
Not diagonal

$$A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \text{ has } A^{\mathrm{T}} = A \text{ and } A^2 = A.$$

When the basis vectors are eigenvectors, the matrix becomes diagonal.

$$\boldsymbol{v}_1 = \boldsymbol{w}_1 = (1, -1) \text{ projects to itself} : T(\boldsymbol{v}_1) = \boldsymbol{v}_1 \text{ and } \lambda_1 = 1$$
$$\boldsymbol{v}_2 = \boldsymbol{w}_2 = (1, \quad 1) \text{ projects to zero } : T(\boldsymbol{v}_2) = \mathbf{0} \text{ and } \lambda_2 = 0$$

Eigenvector bases
Diagonal matrix

The new matrix is $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \Lambda.$ (6)

Eigenvectors are the perfect basis vectors. They produce the eigenvalue matrix Λ.

Sec 8.3

The search for a good basis

Coordinate vectors depend on the choice of basis.

- **Consider two different bases for the same vector space:**

$$S = \{\vec{s}_1, \quad \vec{s}_2, \quad \cdots \quad \vec{s}_n\}, \qquad T = \{\vec{t}_1, \quad \vec{t}_2, \quad \cdots \quad \vec{t}_n\}$$
$$V = span(S) = span(T)$$

$$\forall \vec{x} \in V, \vec{x} = \sum_{i=1}^{n} a_i \vec{s}_i = \sum_{i=1}^{n} b_i \vec{t}_i \,, \rightarrow [\vec{x}]_S = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, [\vec{x}]_T = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

- **How do we change between bases or find a good base?**

Subspaces of $\mathbb{R}^n$.

- **If V is an m-dimensional subspace of $\mathbb{R}^n$, then $\vec{s}_i$ and $\vec{t}_i$ are "traditional" vectors.**

$$\forall \vec{x} \in V \subseteq \mathbb{R}^n, \vec{x} = \sum_{i=1}^{m} a_i \vec{s}_i = \sum_{i=1}^{m} b_i \vec{t}_i, \rightarrow [\vec{x}]_S = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}, [\vec{x}]_T = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$\sum_{i=1}^{m} a_i \vec{s}_i = [\vec{s}_1 \quad \vec{s}_2 \quad \cdots \quad \vec{s}_m] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix} = P[\vec{x}]_S, \qquad \sum_{i=1}^{m} b_i \vec{t}_i = [\vec{t}_1 \quad \vec{t}_2 \quad \cdots \quad \vec{t}_m] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = Q[\vec{x}]_T$$

$$\vec{x} = P[\vec{x}]_S = Q[\vec{x}]_T$$

Subspaces of $\mathbb{R}^n$.

$$\vec{x} = P[\vec{x}]_S = Q[\vec{x}]_T, \qquad P = [\vec{s}_1 \quad \vec{s}_2 \quad \cdots \quad \vec{s}_m], Q = [\vec{t}_1 \quad \vec{t}_2 \quad \cdots \quad \vec{t}_m]$$

- **Both P and Q (n-by-m) have linearly independent columns (full column rank).**
- **Both P and Q (n-by-m) have a left inverse: $P^+P = Q^+Q = I_m$**

$$P[\vec{x}]_S = Q[\vec{x}]_T \rightarrow [\vec{x}]_S = P^+Q[\vec{x}]_T, [\vec{x}]_T = Q^+P[\vec{x}]_S$$

- **There is always a nonsingular (invertible) matrix that can convert between bases (because P and Q have the same column space):**

$$B = Q^+P, B^{-1} = P^+Q \rightarrow [\vec{x}]_T = B[\vec{x}]_S, [\vec{x}]_S = B^{-1}[\vec{x}]_T$$

Change of basis matrix.

$$[\vec{x}]_T = B[\vec{x}]_S, \qquad [\vec{x}]_S = B^{-1}[\vec{x}]_T$$

- The columns of B are just the coordinates of the vectors $S = \{\vec{s}_1, \quad \vec{s}_2, \quad \cdots \quad \vec{s}_n\}$ written relative to $T = \{\vec{t}_1, \quad \vec{t}_2, \quad \cdots \quad \vec{t}_n\}$.
  - The coefficients needed to write: $\vec{s}_i = a_{1i}\vec{t}_1 + a_{2i}\vec{t}_2 + \cdots + a_{ni}\vec{t}_n$

$$B = [[\vec{s}_1]_T \quad [\vec{s}_2]_T \quad \cdots \quad [\vec{s}_n]_T], \qquad B^{-1} = [[\vec{t}_1]_S \quad [\vec{t}_2]_S \quad \cdots \quad [\vec{t}_n]_S],$$

- Why? Because $[\vec{s}_i]_T = B[\vec{s}_i]_S$ and $[\vec{s}_i]_S$ is a vector with a single 1 (and all other elements 0), that "picks out" column i from B!

- Because we are using only coordinate vectors here, the same approach works perfectly well with generalized vectors as well!

Explanation

- $[\vec{s}_1]_S = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$
- $[\vec{s}_1]_T = B[\vec{s}_1]_S = B \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$
- Hence the first column of B = $[\vec{s}_1]_T$

Changing the basis of a transformation matrix.

$$F: V \to U, \qquad [F(\vec{x})] = [F][\vec{x}] \leftrightarrow A\vec{v} = \vec{u}$$

- **If we want to change the basis of for $\vec{v} = [\vec{x}]$, we multiply by the change of basis matrix for V:**

$$\vec{v} = B_V \vec{v}'$$

- **If we want to change the basis of for $\vec{u} = [F(\vec{x})]$, we multiply by the change of basis matrix for U:**

$$\vec{u} = B_U \vec{u}'$$

$$A\vec{v} = \vec{u} \to AB_V\vec{v}' = B_U\vec{u}' \to B_U^{-1}AB_V\vec{v}' = \vec{u}'$$

$$A'\vec{v}' = \vec{u}', A' = B_U^{-1}AB_V$$

Similar matrices.

- When we have a transformation with the same input and output spaces, $F: V \to V$, we may wish to use the same basis on both sides.

$$A'\vec{v}' = \vec{u}', A' = B_U^{-1} A B_V \to A' = B^{-1} A B$$

- When two square matrices share this relationship, we say that they are ***similar***; they represent the same linear transformation, relative to different bases.
- Compare with the eigenvalue decomposition:

$$A = X \Lambda X^{-1} \leftrightarrow \Lambda = X^{-1} A X$$

- The eigenvalue decomposition finds a basis for V (the eigenvector basis) in which the linear transformation is diagonal!
  - Not always possible using a single basis (not all matrices are diagonalizable).

Transformation matrices and the SVD.

- **Moving back to the general case, $F: V \to U$:**

$$A'\vec{v}' = \vec{u}', A' = B_U^{-1} A B_V \to A = B_U A' B_V^{-1}$$

- **Compare with the singular value decomposition:**

$$A = U\Sigma V^T,\ V^T = V^{-1},\ U^T = U^{-1}$$

- **The singular decomposition finds an orthonormal basis for V and a (separate) orthonormal basis for U (the singular vector bases) in which the linear transformation is diagonal.**
- **This can be done for any linear mapping!**

Orthogonality and generalized vector spaces.

- **The SVD gives orthonormal bases for both the input and output spaces (domain and codomain), but what does this mean if we are dealing with generalized vector spaces?**
- **When we begin with a matrix representation of a linear map, everything is already based on coordinate vectors.**
- **One type of orthogonality is the orthogonality of coordinate vectors:**
  - **Coordinate vectors are orthogonal to one another if and only if the dot products of the coordinate vectors are equal to one.**
- **We can apply Gram-Schmidt orthonormalization to coordinate vectors to find an orthonormal set.**

Strang's Book

For different spaces V and W, and different bases v's and w's, we can choose a matrix A related to the bases for a linear mapping $T: V \to W$. Probably A is not symmetric or even square. But we can always choose v's and w's that produce a diagonal matrix using singular value decomposition.

Singular vectors The SVD says that $U^{-1}AV = \Sigma$. The right singular vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ will be the input basis. The left singular vectors $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m$ will be the output basis. By the rule for matrix multiplication, the matrix for the same transformation in these new bases is $B_{\text{out}}^{-1}AB_{\text{in}} = U^{-1}AV = \Sigma$.

Strang's Book

Pure algebra If A is the matrix for a transformation T in the standard basis, then

$$B_{\mathbf{out}}^{-1} A B_{\mathbf{in}} \text{ is the matrix in the new bases.} \tag{1}$$

The standard basis vectors are the *columns of the identity*: $B_{\mathbf{in}} = I_{\mathbf{n}\times\mathbf{n}}$ and $B_{\mathbf{out}} = I_{\mathbf{m}\times\mathbf{m}}$. Now we are choosing special bases to make the matrix clearer and simpler than A. When $B_{\text{in}} = B_{\text{out}} = B$, the square matrix $B^{-1}AB$ is *similar* to A.

1 $B_{\text{in}} = B_{\text{out}} =$ **eigenvector matrix** X. Then $X^{-1}AX =$ **eigenvalues in** Λ.

This choice requires A to be a square matrix with n independent eigenvectors. "A must be diagonalizable." We get Λ when $B_{\text{in}} = B_{\text{out}}$ is the eigenvector matrix X.

2 $B_{\text{in}} = V$ and $B_{\text{out}} = U$: **singular vectors of** A. Then $U^{-1}AV =$ **diagonal** Σ.

Σ is the singular value matrix (with $\sigma_1, \ldots, \sigma_r$ on its diagonal) when B_{in} and B_{out} are the singular vector matrices V and U. Recall that those columns of B_{in} and B_{out} are orthonormal eigenvectors of $A^{\mathrm{T}}A$ and AA^{T}. Then $A = U\Sigma V^{\mathrm{T}}$ gives $\Sigma = U^{-1}AV$.

3 $B_{\text{in}} = B_{\text{out}} =$ **generalized eigenvectors of** A. Then $B^{-1}AB =$ **Jordan form** J.

A is a square matrix but it may only have s independent eigenvectors. (If $s = n$ then B is X and J is Λ.) In all cases Jordan constructed $n - s$ additional "generalized" eigenvectors, aiming to make the Jordan form J *as diagonal as possible*:

i) There are s square blocks along the diagonal of J.

ii) Each block has one eigenvalue λ, one eigenvector, and 1's above the diagonal.

(Jordan form) If A has s independent eigenvectors, it is similar to a matrix J that has s Jordan blocks $J_1 \ldots, J_s$ on its diagonal. Some matrix B puts A into Jordan form:

Jordan form

$$B^{-1}AB = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_s \end{bmatrix} = J. \tag{3}$$

Each block J_i has one eigenvalue λ_i, one eigenvector, and 1's just above the diagonal:

Jordan block

$$J_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \cdot & \cdot & \\ & & \cdot & 1 \\ & & & \lambda_i \end{bmatrix}. \tag{4}$$

Matrices are similar if they share the same Jordan form J—not otherwise.

The Jordan form solves the differential equation $du/dt = Au$ for **any square matrix** $\boldsymbol{A = BJB^{-1}}$. The solution $e^{At}\boldsymbol{u}(0)$ becomes $\boldsymbol{u}(t) = Be^{Jt}B^{-1}\boldsymbol{u}(0)$. J is triangular and its matrix exponential e^{Jt} involves $e^{\lambda t}$ times powers $1, t, \ldots, t^{s-1}$.

Orthogonality and generalized vector spaces.

- **We have seen that we can write coordinate vectors for any vector space, so long as we can define a basis.**
- **One type of orthogonality is the orthogonality of coordinate vectors:**
  - **Coordinate vectors are orthogonal to one another if and only if the dot products of the coordinate vectors are equal to zero.**
- **Is there another kind of orthogonality?**
  - **Orthogonal vectors have no components "in the same direction" as the others; the projection of orthogonal vectors onto one other is the zero vector.**
  - **This greatly simplifies how we find a linear combination of basis vectors:**

$$\vec{u} = \sum_{i=1}^{n} a_i \vec{q}_i \rightarrow a_i = \frac{\vec{u} \cdot \vec{q}_i}{\vec{q}_i \cdot \vec{q}_i}$$

- **Can we define an orthogonal basis for the underlying vector space?**

Orthogonality of the underlying basis vectors.

We can discuss orthogonality of generalized vectors in a deeper sense by introducing the concept of a generalized inner product

$$\langle \vec{u}, \vec{v} \rangle$$

- Just like the dot product:
  - **Take two vectors as inputs and returns a scalar as an output.**
  - **Is (conjugate) symmetric:**

$\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$ **if real;** $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle^*$ **if complex**

  - **Is positive definite:**

$$\langle \vec{u}, \vec{u} \rangle > 0, \forall \vec{u} \neq \vec{0}$$

- **Any function with these properties can be considered an inner product!**

An inner product for functions.

- **A commonly used inner product for function space is the integral of the product of the two functions over the functions' domain:**

$$\langle f(x), g(x) \rangle = \int_a^b f(x)g(x)dx$$

- **Just like the dot product takes the sum of corresponding entries in a ordered list, this definition takes the sum of matched infinitesimals (extremely small)!**

The naïve basis for polynomial space is not orthogonal!

- **Consider the simple but terrible basis for functions:**

$$\{1, \quad x, \quad x^2, \quad x^3, \quad \cdots\}$$

- **Two common domains: [0,1] and [-1,1]**

  Any domain of [0,b] is easily mapped onto the first, and any domain of [-b,+b] is easily mapped onto the second.

- **Even and odd powers are orthogonal to one another on the second domain, but not on the first; evens are never orthogonal to other evens and odds are never orthogonal to odds.**

$$\langle 1, x \rangle = \int_{-1}^{1} x dx = \frac{1}{2} x^2 \Big|_{-1}^{1} = 0, \qquad \langle 1, x \rangle = \int_{0}^{1} x dx = \frac{1}{2} x^2 \Big|_{0}^{1} = \frac{1}{2}$$

$$\langle 1, x^2 \rangle = \int_{-1}^{1} x^2 dx = \frac{1}{3} x^3 \Big|_{-1}^{1} = \frac{2}{3}, \qquad \langle x, x^3 \rangle = \int_{-1}^{1} x^4 dx = \frac{1}{5} x^5 \Big|_{-1}^{1} = \frac{2}{5}$$

$$\begin{aligned}
\text{Inner product } (\boldsymbol{f}, \boldsymbol{g}) &= \int f(x) g(x)\, dx \\
\text{Complex inner product } (\boldsymbol{f}, \boldsymbol{g}) &= \int \overline{f(x)}\, g(x)\, dx, \ \ \overline{f} = \text{ complex conjugate} \\
\text{Weighted inner product } (\boldsymbol{f}, \boldsymbol{g})_w &= \int w(x)\, \overline{f(x)}\, g(x)\, dx, \ \ w = \text{ weight function}
\end{aligned}$$

When the integrals go from $x = 0$ to $x = 1$, the inner product of x^i with x^j is

$$\int_0^1 x^i\, x^j\, dx = \left.\frac{x^{i+j+1}}{i+j+1}\right]_{x=0}^{x=1} = \frac{1}{i+j+1} = \text{ entries of Hilbert matrix } B^{\mathrm{T}} B$$

By changing to the symmetric interval from $x = -1$ to $x = 1$, we immediately have *orthogonality between all even functions and all odd functions*:

Interval $[-1,\ 1]$ $\qquad \displaystyle\int_{-1}^{1} x^2\, x^5\, dx = 0 \qquad \int_{-1}^{1} \mathbf{even}(x)\, \mathbf{odd}(x)\, dx = 0.$

How could we create an orthogonal basis?

Apply Gram-Schmidt with ANY inner product!

- **General rule:**

$$\vec{w}_i = \vec{a}_i - \sum_{j=1}^{i-1} \frac{\langle \vec{a}_i, \vec{w}_j \rangle}{\langle \vec{w}_j, \vec{w}_j \rangle} \vec{w}_j \,, \qquad \vec{q}_i = \frac{1}{\|\vec{w}_i\|} \vec{w}_i, \|\vec{w}_i\| = \sqrt{\langle \vec{w}_i, \vec{w}_i \rangle}$$

- **Let's orthogonalize the naïve basis, using the domain of [-1,1]:**

$$\{1, \quad x, \quad x^2, \quad x^3, \quad \cdots\}$$

$$\vec{w}_1 = 1, \qquad \langle \vec{w}_1, \vec{w}_1 \rangle = \int_{-1}^{1} 1 dx = x\Big|_{-1}^{1} = 2 \rightarrow \vec{q}_1 = \frac{1}{\sqrt{2}}$$

$$\vec{w}_2 = x - \frac{\langle x, 1 \rangle}{\langle 1,1 \rangle} 1 = x, \qquad \langle x, 1 \rangle = \int_{-1}^{1} x dx = \frac{1}{2} x^2 \Big|_{-1}^{1} = 0$$

$$\langle \vec{w}_2, \vec{w}_2 \rangle = \int_{-1}^{1} x^2 dx = \frac{1}{3} x^3 \Big|_{-1}^{1} = \frac{2}{3} \rightarrow \vec{q}_2 = \frac{\sqrt{3}}{\sqrt{2}} x$$

Apply Gram-Schmidt with ANY inner product!

$$\vec{w}_1 = 1, \vec{w}_2 = x, \qquad \vec{q}_1 = \frac{1}{\sqrt{2}}, \vec{q}_2 = \frac{\sqrt{3}}{\sqrt{2}}x$$

$$\vec{w}_3 = x^2 - \frac{\langle x^2, 1\rangle}{\langle 1,1\rangle}1 - \frac{\langle x^2, x\rangle}{\langle x, x\rangle}x = x^2 - \frac{1}{3}$$

$$\langle x^2, 1\rangle = \int_{-1}^{1} x^2 dx = \frac{1}{3}x^3\Big|_{-1}^{1} = \frac{2}{3}, \qquad \langle x^2, x\rangle = \int_{-1}^{1} x^3 dx = \frac{1}{4}x^4\Big|_{-1}^{1} = 0$$

$$\langle \vec{w}_3, \vec{w}_3\rangle = \int_{-1}^{1}\left(x^2 - \frac{1}{3}\right)^2 dx = \frac{1}{5}x^5 - \frac{2}{9}x^3 + \frac{1}{9}x\Big|_{-1}^{1} = \frac{8}{45} \rightarrow \vec{q}_3 = \frac{3\sqrt{5}}{2\sqrt{2}}\left(x^2 - \frac{1}{3}\right)$$

Apply Gram-Schmidt with ANY inner product!

$$\vec{w}_1 = 1, \vec{w}_2 = x, \vec{w}_3 = x^2 - \frac{1}{3}, \qquad \vec{q}_1 = \frac{1}{\sqrt{2}}, \vec{q}_2 = \frac{\sqrt{3}}{\sqrt{2}}x, \vec{q}_3 = \frac{3\sqrt{5}}{2\sqrt{2}}\left(x^2 - \frac{1}{3}\right)$$

$$\vec{w}_4 = x^3 - \frac{\langle x^3, 1\rangle}{\langle 1,1\rangle}1 - \frac{\langle x^3, x\rangle}{\langle x, x\rangle}x - \frac{\left\langle x^3, x^2 - \frac{1}{3}\right\rangle}{\left\langle x^2 - \frac{1}{3}, x^2 - \frac{1}{3}\right\rangle}\left(x^2 - \frac{1}{3}\right) = x^3 - \frac{3}{5}x$$

$$\langle x^3, 1\rangle = \left\langle x^3, x^2 - \frac{1}{3}\right\rangle = 0, \qquad \langle x^3, x\rangle = \int_{-1}^{1} x^4 dx = \frac{1}{5}x^5\Big|_{-1}^{1} = \frac{2}{5}$$

$$\langle \vec{w}_4, \vec{w}_4\rangle = \int_{-1}^{1}\left(x^3 - \frac{3}{5}x\right)^2 dx = \frac{1}{7}x^7 - \frac{6}{25}x^5 + \frac{3}{25}x^3\Big|_{-1}^{1} = \frac{8}{175} \rightarrow \vec{q}_4 = \frac{5\sqrt{7}}{2\sqrt{2}}\left(x^3 - \frac{3}{5}x\right)$$

The Legendre Polynomials- Orthogonal Basis:

- **Unnormalized:**

$$L = \left\{1, \quad x, \quad x^2 - \frac{1}{3}, \quad x^3 - \frac{3}{5}x, \quad \cdots\right\}$$

- **Normalized**

$$L = \left\{\frac{1}{\sqrt{2}}, \quad \frac{\sqrt{3}}{\sqrt{2}}x, \quad \frac{3\sqrt{5}}{2\sqrt{2}}\left(x^2 - \frac{1}{3}\right), \quad \frac{5\sqrt{7}}{2\sqrt{2}}\left(x^3 - \frac{3}{5}x\right), \quad \cdots\right\}$$

The Fourier Basis is “Naturally” Orthogonal!

- **The Fourier Basis is defined in terms of sines and cosines**

$$F = \{1, \quad \sin(x), \quad \cos(x), \quad \sin(2x), \quad \cos(2x), \quad \cdots\}$$

- **This basis is naturally orthogonal on any domain of** $[\mathrm{a}, \mathrm{a} + 2\pi]$*:*

$$\langle f(x), g(x) \rangle = \int_0^{2\pi} f(x) g(x) dx$$

Examples for Orthogonal bases for function spaces

5. The **Fourier basis**	$1, \sin x, \cos x, \sin 2x, \cos 2x, \ldots$
6. The **Legendre basis**	$1, x, x^2 - \frac{1}{3}, x^3 - \frac{3}{5}x, \ldots$
7. The **Chebyshev basis**	$1, x, 2x^2 - 1, 4x^3 - 3x, \ldots$

Fourier Series

The *Fourier transform* connects $f(x)$ to the coefficients a_k and b_k in its Fourier series:

Fourier series $$f(x) = a_0 + b_1 \sin x + a_1 \cos x + b_2 \sin 2x + a_2 \cos 2x + \cdots$$

Example 3 The double angle formula in trigonometry is $\cos 2x = 2\cos^2 x - 1$. This tells us that $\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$. A very short Fourier series. So is $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$.